INDIAN SCHOOL MUSCAT

SENIOR SECTION

HOLIDAY HOME WORK-2026

CLASS XI

SCIENCE STREAM

Persistence is important in every endeavor. Whether it's finishing your homework, completing school, working late to finish a project, or "finishing the drill" in sports, winners persist to the point of sacrifice in order to achieve their goals.

Leon F. "Lee" Ellis

Class 11- ALS (Assessment of Listening and Speaking) PROJECT PORTFOLIO

The Rough draft- 29th July, 2026

Final Submission- 2nd August 2026.

Select topics for the project:

Interview – Students can choose a topic for their interview. Prepare a questionnaire, meet several neighbours belonging to various professional fields and prepare a report (800- 1000 words)

Documentaries: Listen, watch a documentary on a topic that may be educational, entertainment – music, dance, short films, etc., and prepare a report (800- 1000 words). **(Topic to be approved by the teacher in charge)**

Specifications: 10 MARKS (5 for report + 5 for Viva-Voce)

How to prepare: **(THE PROJECT NEEDS TO BE HAND-WRITTEN)**

Cover Page : Use the cover page uploaded to Moodle to ensure uniformity.

Title page: **Page 1:** Top Centre – Name of the School,

Centre Below - ALS Project Portfolio

Followed by - Topic of your Project (Topic Sentence).

Full Name,

Class

Section

Roll No.

Can be a colourful title page, with a suitable quote related to your topic, at the extreme bottom of all the above-mentioned details.

Page 2: The main Objectives of the Project: a brief description of the purpose, procedure and conduct of your project – approximately 100 words.

Page 3: centre

Page 4: **Top Centre:** **ACTION PLAN**

(about 50 words - State how the work was done, simple mention of the sources consulted – experts, peers, professionals, print and visual media, etc.)

Page 5: Material Evidences (e.g.: attach questionnaires, may be a set of 10-15 simple questions related to your topic In case of documentary – comments by critics, friends, reviews by renowned individuals)

Page 6: Media Material (mention the media you used – attach a screen print of it) / Books and Journals you referred: Title of the journal/ book/s, Name of the Author (attach the screen print/picture of the cover of the book/ journal)

Page 7: Project Report

This is the main part of the project work. It may run up to 4-5 pages (600 words), neatly paragraphed pages. All aspects of the project work are explained in a systematic manner.

Page 8:– Conclusion (one paragraph, roughly 50 words) – the end of the project report –

Page 9: Student/ group reflections: Your learning/How did you benefit / What next? (50 words)

Prepare the report, keep the record of the project and prepare for Viva.

Attach one or two photographs of learning experiences (Images of interview / in case of documentary – prominent scenes, etc)

Page 10: List of resources or bibliography

Last Page: Thanks for the review. (50 words)

Thank everyone who gave you the opportunity and all those who helped you.

The **following points will be kept for consideration while assessing the project portfolios:**

- Quality of content of the project
- Accuracy of information
- Adherence to the specified timeline
- Content in respect of (spellings, grammar, punctuation)
- Clarity of thoughts and ideas
- Creativity/ presentation & Knowledge and experience gained

MARKING OF PROJECT FILE: 5m Content Coverage (2mark), Presentation (1Mark), Completeness with grammatical accuracy (1mark), Quality (1mark)

Ms. Novena Augustine

Head of the Department

Certificate of Completion.

This project work titled…..…….. is as per Class 11 CBSE guidelines and has been carried out with the guidance and instructions of Mr./ Ms…………………….
…………….PGT English of Indian School Muscat.

Signature – Mentor and Guide

INDIAN SCHOOL MUSCAT
HOLIDAY HOMEWORK
2026-27
CLASS XI_: MATHEMATICS

Instructions: -

Dear student,

A. Write the following activities and draw the related figures in your Mathematics activity notebook. Please refer to your Moodle classrooms for the write ups.

ACT. No.	ACTIVITY
1.	Number of subsets of a set
2.	Distinguish between relations and functions
3.	Prepare a model to illustrate the values of sine function and cosine function for different angles which are multiples of pi/2 and pi

B. Use A4 size, ruled sheets, to solve the following questions.

QUESTION ANSWERS

In each of the following questions, a statement of **Assertion (A)** is followed by a statement of **Reason (R)**. Mark the correct choice as:

- **(a)** Both (A) and (R) are true and (R) is the correct explanation of (A).
- **(b)** Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
- **(c)** (A) is true but (R) is false.
- **(d)** (A) is false but (R) is true.

1.	**Assertion (A):** If a set X has 5 elements, then its power set $P(X)$ contains 32 elements. **Reason (R):** The total number of proper subsets of a set with n elements is given by $2^n - 1$.
2.	**Assertion (A):** The graph of the signum function changes values sharply, but its range consists of only three distinct elements: $\{-1, 0, 1\}$. **Reason (R):** The signum function yields 1 for all positive inputs, -1 for all negative inputs, and 0 when the input is exactly zero.

3.	**Digital Streaming Ecosystems** **Scenario:** A tech research firm surveyed **120 working professionals** to map out digital subscription overlapping behaviors across three major ecosystems: **Netflix (N)**, **Amazon Prime (P)**, and **Hotstar (H)**. The structured data metrics highlighted the following distribution: • **65 professionals** watch Netflix. • **50 professionals** watch Amazon Prime. • **55 professionals** watch Hotstar. • **25 professionals** maintain dual subscriptions to Netflix and Amazon Prime ($N \cap P$). • **20 professionals** maintain dual subscriptions to Amazon Prime and Hotstar ($P \cap H$). • **22 professionals** maintain dual subscriptions to Netflix and Hotstar ($N \cap H$). • **10 professionals** are power-users holding subscriptions to **all three** premium services ($N \cap P \cap H$). **Questions:** 1. Draw or list the numeric values for a 3-circle Venn diagram layout, showing the number of individuals in the **central shared core intersection** ($N \cap P \cap H$) and the three **exclusive dual intersections** (e.g., Netflix and Prime *only*). 2. Find the exact number of professionals who subscribe to **Netflix only.** 3. Calculate the number of surveyed individuals who subscribe to **exactly two** streaming platforms. 4. Determine how many professionals do **not subscribe to any** of these three entertainment services.
4.	**Case Study Based Question** **Tidal Wave Modelling** Oceanographers monitoring the water depth at a harbor inlet model the fluctuating water level using a periodic trigonometric function. The height $H(t)$ (in meters) of the tide above the sea floor is modeled as a function of time t (in hours past midnight) by the following continuous structural layout:

```
Water Height H(t)

    ▲
12 ┤       • (3, 12) [High Tide]
10 ┤      / \
 8 ┤     /   \
 6 ┤  •       • (9, 6) [Low Tide]
   ┼──┴──┬──┬──┬──┬──► Time t (Hours past midnight)

   0     3  6  9  12
```

The mathematical equation representing this curve is given by:

$$H(t) = 9 + 3\cos\left(\frac{\pi t}{6}\right)$$

Based on the tracking data and formula provided, solve the following sub-questions:

1. State the **maximum** and **minimum** heights of the water at the harbor.
2. Find the exact height of the water at **3:00 AM** ($t = 3$).
3. Determine the water height at **8:00 AM** ($t = 8$). Keep your answer in decimal form.
4. Find the first two positive values of time t when the water depth is **exactly 9 meters.**

5.

Municipal Water Billing (Piecewise Functions)

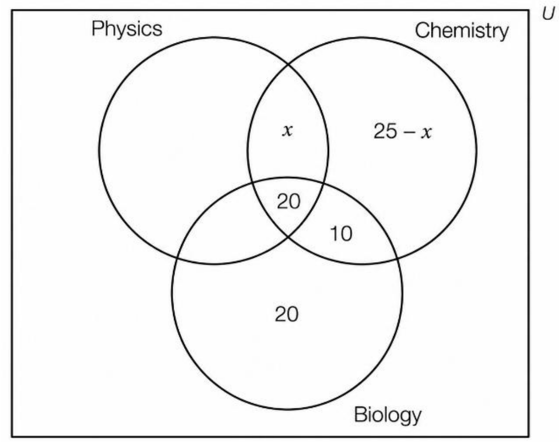
Scenario:
To encourage water conservation, a municipal corporation charges residential societies for water consumption using a tiered, piecewise-defined pricing function $W(x)$. Here, x represents the volume of water consumed in kiloliters (kL) per month, and $W(x)$ is the total monthly bill in ₹.

The billing structure is defined as follows:

- Consuming up to 15 kL costs a flat baseline rate of ₹300.
- Consuming more than 15 kL costs the baseline rate of ₹300 plus an additional volumetric fee of ₹20 for every kL consumed *above* the initial 15 kL limit.

Questions:

1. Write down the complete algebraic piecewise formula for $W(x)$ for all $x \geq 0$.
2. Calculate the exact water bill amount for a household that consumes **27 kL** of water in a month.
3. State the continuous **Domain** and **Range** of this water billing function based on the municipal parameters.

6.	At a certain school there are 90 students studying for their IB diploma. They are required to study at least one of the subjects: Physics, Biology or Chemistry. • 50 students are studying Physics, • 60 students are studying Biology, • 55 students are studying Chemistry, • 30 students are studying both Physics and Biology, • 10 students are studying both Biology and Chemistry but not Physics, • 20 students are studying all three subjects.  Let x represent the number of students who study both Physics and Chemistry but not Biology, then $25 - x$ is the number who study Chemistry only. The figure alongside shows some of this information and can be used for working. **(a)** Express the number of students who study Physics only, in terms of x. **(b)** Find x. **(c)** Determine the number of students studying at least two of the subjects

INDIAN SCHOOL MUSCAT
DEPARTMENT OF PHYSICS
CLASS- XI 2026-27

HOLIDAY HOME WORK

I. MULTIPLE CHOICE QUESTIONS

1. In the C.G.S. system the magnitude of the force is 100 dyne. In another system where the fundamental physical quantities are kilogram, metre, and minute, the magnitude of the force is:

   (a) 0.036 (b) 0.36 (c) 3.6 (d) 36

2. If momentum (P), area (A) and time (T) are taken to be fundamental quantities, then energy has the dimensional formula:

   (a) $(P^1 A^{-1} T^1)$ (b) $(P^2 A^1 T^1)$ (c) $(P^1 A^{-1/2} T^1)$ (d) $(P^1 A^{1/2} T^{-1})$

3. If $x = a + bt + ct^2$, where x is in metre and t in second, then what is the unit of 'c'?

   (a) m/s (b) m/s^2 (c) kgm/s (d) m^2/s

4. A particle is moving with a constant speed along a straight-line path. A force is not required to

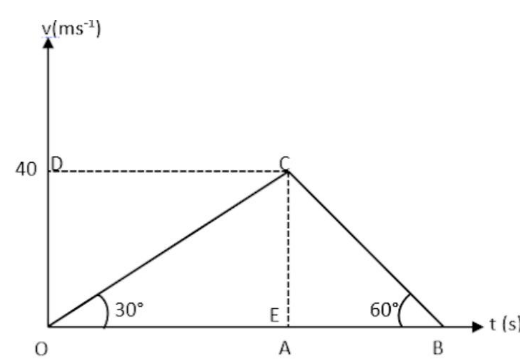
   (a) change its direction (b) decrease its speed

   (c) keep it moving with uniform velocity (d) Increase its momentum

5. A body is dropped from rest from a height h. It covers a distance 9h/25 in the last second of fall. The height h is ($g = 10\ m/s^2$)

   (a) 102.5 m (b) 112.5 m (c) 122.5 m (d) 132.5 m

6. The velocity-time graph of a body is shown in figure, for the interval OA and AB the ratio of distance covered is:



(a) 3:1 (b) 1:3 (c) √3:1 (d) √3:2

7. A spring with one end attached to a mass and the other to a rigid support is stretched and released.

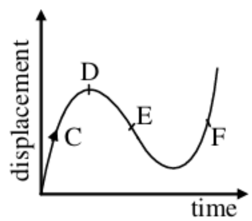
(a) Magnitude of acceleration, when just released is maximum.

(b) Magnitude of acceleration, when at equilibrium position, is maximum.

(c) Speed is maximum when mass is at equilibrium position.

(d) Magnitude of displacement is always maximum whenever speed is minimum.

8. The displacement-time graph of a moving particle is shown. The instantaneous velocity of the particle is negative at the point



(a)D (b) F (c) C (d) E

9. The displacement of a particle is represented by the following equation: $s = 3t^3 + 7t^2 + 5t + 8$ where s is in metre and t in second. The acceleration of the particle at t = 1 is

(a) 14 m/s^2 (b) 18 m/s^2 (c) 32 m/s^2 (d) zero

II. ASSERTION -- REASON QUESTIONS

Two statements are given: one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer to these questions from the codes (a), (b), (c) and (d) as given below.

a) Both A and R are true and R is the correct explanation of A

b) Both A and R are true and R is NOT the correct explanation of A

c) A is true but R is false

d) A is false and R is also false

10. Assertion: Radian is the SI unit of plane angle.
Reason: One radian is the angle subtended at the centre of the circle by an arc whose length is equal to the radius of the circle.

11. Assertion: Sum of 7.21, 12.141 and 0.0028 is 19.35.

Reason: Significant figures in the sum or difference of two numbers should be reported with the same number of decimal places as that of the number with minimum number of decimal places.

12. Assertion: Special functions such as trigonometric, logarithmic and exponential functions are not dimensionless.

    Reason: A pure number, ratio of similar physical quantities such as angle and refractive index has some dimensions.

13. Assertion: A particle having constant acceleration must always move on a straight line.

    Reason: When magnitude of acceleration is constant, then speed of particle may remain constant.

14. Assertion: The position-time graph of a uniform motion, in one dimension of a body cannot have negative slope.

    Reason: In one-dimensional motion the position does not reverse, so it cannot have a negative slope.

15. Assertion: For the uniform motion, the slope of position time graph will be constant.

    Reason: The slope of position time graph represents velocity of the object and for uniform motion it is constant

16. Assertion: Two particles of different mass, projected with same velocity at same angles. The maximum height attained by both the particle will be same.

    Reason: The maximum height of projectile is independent of particle mass.

17. Assertion**:** In projectile motion the vertical velocity of the particle is continuously decreased during its ascending motion.

    Reason: In projectile motion downward constant acceleration is present in vertical direction.

III. CASE BASED QUESTION

18. Every measurement involves errors. Thus, the result of measurement should be reported in a way that indicates the precision of measurement. Normally, the reported result of measurement is a number that includes all digits in the number that are known reliably plus the first digit that is uncertain. The reliable digits plus the first uncertain digit are known as significant digits or significant figures.

    i) If the length and breadth are measured as 4.234 and 1.05 m, the area of the rectangle is

    (a) 4.4457 m^2 (b) 4.45 m^2 (c) 4.446 m^2 (d) 0.4446 m^2

    ii) Which of the following has least significant figures?

    (a) 0.80760 (b) 0.80200 (c) 0.08076 (d) 80.267

iii) How many significant figures does 190909090?

(a) 8 (b) 7 (c) 9 (d) 5

iv) Express the final answer up to proper significant digits: 101.2 + 18.702 =?

(a) 119.902 (b) 119.9 (c) 119.90 (d) 119.91

OR

How many significant figures are in measurement 0.0082 L?

(a) 2 (b) 4 (c) 5 (d) 3

19. When an object moves along a straight line with uniform acceleration, it is possible to relate its velocity, acceleration during motion and the distance covered by it in a certain time interval by a set of equations known as the equations of motion. For convenience, a set of three such equations are given below:

$v = u + at$

$s = ut + \frac{1}{2} at^2$

$2a\,s = v^2 - u^2$ where u is the initial velocity of the object which moves with uniform acceleration a for time t, v is the final velocity and s is the distance travelled by the object in time t.

i) Equations of motion are applicable to motion with

(a) uniform acceleration

(b) non uniform acceleration

(c)constant velocity

(d) none of these

ii) Consider a body moving with an acceleration of 2 m/s^2. After t seconds its velocity is 10 m/s. Find 't'.

(a) 4 s (b) 20 s (c) 5 s (d) 8s

iii) The brakes applied to a car produce an acceleration of 10 ms^{-2} in the opposite direction to the motion. If the car takes 1 s to stop after the application of brakes, calculate the distance covered during this time by car.

(a) 5m (b) 2.5m (c)25m D) 10m

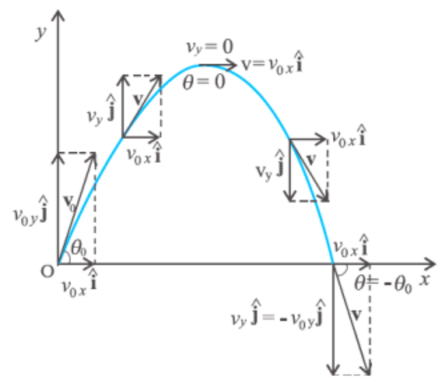
iv) An object is dropped from a tower falls with a constant acceleration of 10 m/s^2 . Find its speed 10 s after it was dropped.

(a)110m/s (b)100m/s (c)120m/s (d) -120m/s OR

A bullet hits a Sand box with a velocity of 10 m/s and penetrates it up to a distance of 5 cm. Find the deceleration of the bullet in the sand box.

(a) − 10m⁄s (b) 20m⁄s (c) $-9 m/s^2$ (d)$-18 m/s^2$

20. An object that is in flight after being thrown or projected is called a projectile. Such a projectile might be a football, a cricket ball, a baseball or any other object. The motion of a projectile may be thought of as the result of two separate, simultaneously occurring components of motions. One component is along a horizontal direction without any acceleration and the other along the vertical direction with constant acceleration due to the force of gravity. It was Galileo who first stated this independency of the horizontal and the vertical components of projectile motion in his Dialogue on the great world systems (1632). In our discussion, we shall assume that the air resistance has negligible effect on the motion of the projectile. Suppose that the projectile is launched with velocity vo that makes an angle θo with the x-axis as shown in Fig. After the object has been projected, the acceleration acting on it is that due to gravity which is directed vertically downward.



i) Out of which of the following cases which can be called a projectile motion?

(a)A child takes a turn on the ground

(b)A player throws a ball above a wall

(c)A block slides on floor

(d) A car got a collision with other car

ii)At which point of the path it's angle of velocity from the horizontal becomes 0 ^{o}?

(a) At point O (b) At top point

(c) 1 sec after projection (d) 2 sec after projection

iii) If the velocity of projection of a projectile is $2\hat{i} + 3\hat{j}$, then the velocity with which it strikes the ground is:

(a) $-2\hat{\imath} + 3\hat{\jmath}$ (b) $-2\hat{\imath} - 3\hat{\jmath}$ (c) $2\hat{\imath} - 3\hat{\jmath}$ (d) $2\hat{\imath} + 3\hat{\jmath}$

iv)The horizontal component of velocity

(a) goes on decreasing (b) goes on increasing

(c) cannot be predicted (d) remains constant

OR

On increasing angle from the horizontal

(a)Vertical component of velocity increases

(b)horizontal component of velocity increases

(c)No change occurs in components of velocity

(d)Vertical component of velocity decreases

INDIAN SCHOOL MUSCAT

CHEMISTRY

CLASS: XI

HOLIDAY ASSIGNMENT

I	***MULTIPLE CHOICE QUESTION***
1	Number of molecules present in 1ml of water vapor is (a) 1 (c) 1000 (b) 2.69×10^{19} (d) 6.02×10^{20}
2	An organic compound contains carbon, hydrogen and oxygen. Its elemental analysis gave C – 38.71% and H – 9.67%. The empirical formula of the compound would be a) CHO b) CH_4O c) CH_3O d) CH_2O
3	How many grams of carbon dioxide will be formed when 1 mole of carbon is burnt in 16g of dioxygen? (a) 22g (b) 44g (c) 12g (d) 32g
4	Which of the following is dependent on temperature? (a) Mole fraction (b) Molality (c) Molarity (d) Mass percentage
5	A solution is prepared by adding 2 g of NaCl to 18 g of water. The mass percent of the solute is……………. (a) 20 (b) 25 (c) 40 (d) 10
6	If the concentration of glucose ($C_6H_{12}O_6$) in blood is 0.9g/L, what will be the molarity of glucose in blood? (a) 5M (c) 50M (b) 0.005M (d) 0.5M
7	2g of oxygen contain number of atoms equal to that contained by a) 0.5g hydrogen (c) 4g Sulphur b) 7g nitrogen (d)2.3g Sodium
8	According to Bohr's theory, the angular momentum of an electron in 5th orbit is (a) $1\ h/2\pi$ (b) $10h/\pi$ (c) $2.5h/\pi$ (d) $25h/\pi$
9	Calculate the energy in Joule corresponding to light of wavelength 45nm. (a) 6.67×10^{15} (c) 4.42×10^{-15} (b) 6.67×10^{11} (d) 4.42×10^{-18}
10	Which of the following has a minimum wavelength? (a) Gamma rays (b) Blue light (c) Infrared rays (d) microwave
11	The electronic transition from n= 2 to n= 1 will produce the shortest wavelength in (where n = principal quantum number)

	(a) He^+ (b) H (c)H^+ (d) Li^{2+}
12	The ionization enthalpy of hydrogen atom is 1.312 x 10^6 J/mol. The energy required to excite the electron in the atom from n_1= 1 to n_2 = 2 (a) 6.56 x 10^5 J/mol (c) 7.56 x 105 J/mol (b) 9.84 x 10^5 J/mol (d) 8.51 x 10^5 J/mol
II	***ASSERTION REASON QUESTION*** (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true. (e) Both A and R are false.
1	Assertion: Hydrogen has one electron in its orbit but it produces several spectral lines Reason: There are many excited energy levels available.
2	Assertion (A): It is impossible to determine the exact position and exact momentum of an electron simultaneously. Reason (R): The path of an electron in an atom is clearly defined.
3	Assertion (A): Combustion of 16g of methane gives 18g of water Reason (R): In the combustion of methane, water is one of the products
4	Assertion (A): The empirical mass of ethane is half of its molecular mass Reason (R): The empirical formula represent the simplest whole number ratio of various atoms present in a compound
III	***DESCRIPTIVE TYPE QUESTIONS***
1	Calcium carbonate reacts with aqueous HCl to give $CaCl_2$ and CO_2 according to the reaction given below: $CaCO_3$ (s) + 2HCl (aq) → $CaCl_2$(aq) + CO_2(g) + H_2O(l) . What mass of $CaCl_2$ will be formed when 250 mL of 0.76 M HCl reacts with 1000 g of $CaCO_3$? Name the limiting reagent. Calculate the number of moles of $CaCl_2$ formed in the reaction.
2	How many grams of chlorine (Cl_2) are required to completely react with 0.4g hydrogen (H_2) to yield hydrochloric acid? Also calculate the amount of HCl formed.
3	Calculate the wave number of the line having frequency 6 x 10^{16} Hz.
4	Calculate the number of chloride ions in 100ml of 0.01M $AlCl_3$ solution.
5	A compound contains 4.07% hydrogen, 24.27 % carbon and 71.26% chlorine. Its molar mass is 98.96g. What are its empirical and molecular formula? [C =12 u , H = 1u, Cl = 35.5u]
6	Calculate the frequency and the wavelength of the radiation in nanometers emitted when an electron in the hydrogen atom jumps from third orbit to the ground state. In which region of the electromagnetic spectrum will the line lie?
7	A proton is moving with Kinetic energy 5 x 10^{27} J. What is the velocity of the proton?
8	Light of wavelength 1281.8 nm is emitted when an electron of H atom drops from 5th to 3rd energy level. Calculate the wavelength of the photon emitted when electron falls from third to ground level.
9	Calculate number of moles in 8.0 g of O_2. (Atomic mass of O = 16u)

10	An impure sample of NaCl which weigh 1.2g gave on treatment with excess of $AgNO_3$ solution 2.4g of AgCl as a precipitate. Calculate the percentage purity of the sample
11	Calculate the percent of carbon, hydrogen and oxygen in ethanol (C_2H_5OH). (Atomic mass of C = 12u, H = 1u and O =16u)
12	What is limiting reagent? 250 ml of 0.2M $AgNO_3$ solution is added to 400 ml of 0.3M Na_2SO_4 solution resulting in the formation of a white precipitate of silver sulphate. How many moles of silver sulphate will be precipitated? Which is the limiting reagent? (Ag = 108u, S=32u, O=16u)
13	(i) Calculate the molecular mass of sodium sulphate (Na_2SO_4). (Atomic mass of Na = 23, S = 32 ,0 =16) (ii) A bottle contains 500ml of 2.4M HCl solution. How much water should be added to dilute it to 1.6M HCl solution?
14	A solution contains 25% water, 25% ethanol (CH_3CH_2OH) and 50% acetic acid (CH_3COOH) by mass. Calculate the mole fraction of water present in the solution.
15	2.82g of glucose ($C_6H_{12}O_6$) are dissolved in 30g of water. Calculate the molality of the solution.
16	Calculate the mass of sodium acetate (CH_3COONa) required to make 500 mL of 0.375 molar aqueous solution. Molar mass of sodium acetate is 82.0245 g mol^{-1}.
17	A sugar syrup of weight 214.2g contains 34.2 g 0f sugar($C_{12}H_{22}O_{11}$).Calculate (i) Molal concentration (ii) Mole fraction of sugar in the syrup
18	The density of 3 molal solution of NaOH is 1.110g/ml. Calculate the molarity of the solution.
19	What mass of calcium oxide will be obtained by heating 3mol of $CaCO_3$? $CaCO_3 \rightarrow CaO + CO_2$
20	Calculate the wavelength of the spectral line obtained in the spectrum of Li^{2+} ion when the transition takes place between two levels whose sum is 4 and difference is 2.

INDIAN SCHOOL MUSCAT
DEPARTMENT OF BIOLOGY
HOLIDAY HOMEWORK
CLASS XI

NOTE: Write the answers to this questions in your notebook and submit to your teacher by 20th July 2026.

<u>Answer the following questions:</u>

1. Define vital capacity. What is its significance?
2. Which part of the heart is responsible for initiating and maintaining its rhythmic activity?
3. Where does the selective reabsorption of Glomerular filtrate take place?
4. A fluid-filled double membranous layer surrounds the lungs. Name it and mention its important function.
5. Write a note on the mechanism of breathing.
6. Given below are the abnormal conditions related to blood circulation. Name the disorders.
   a. Acute chest pain due to failure of oxygen supply to heart muscles.
   b. Increased systolic pressure.
7. Describe the structure of the human kidney with a labeled diagram.
8. What are the major transport mechanisms for CO2? Explain.
9. Write the differences between blood and lymph.
10. What are the main processes of urine formation?
11. The incidence of emphysema, a respiratory disorder, is high in cigarette smokers. In such cases:
    i. the bronchioles are found damaged.
    ii. the alveolar walls are found damaged.
    iii. the plasma membrane is found damaged.
    iv. the respiratory muscles are found damaged.
12. Thrombocytes are essential for the coagulation of blood. Explain.
13. State the functions of the following in blood.
    a. Fibrinogen b. Globulin c. Neutrophils d. Lymphocytes
14. Why is a hemodialysis unit called an artificial kidney?
15. What is the role played by Renin-Angiotensin in the regulation of kidney function?

……………………………………………………………………………………………

INDIAN SCHOOL MUSCAT
SENIOR SECTION
DEPARTMENT OF ENGINEERING GRAPHICS
CLASS XI
TOPIC – HOLIDAY ASSIGNMENT

The following are to be completed in A4 SHEET FILE Drawing notebook and submit on 19-07-2026

1. Draw a triangle ABC with AB=40 mm, BC=50 mm and CA=60 mm. Draw a circle passing through A, B and C.
2. To draw a line inclined at 15° to a given horizontal line from a given point using set square.
3. Draw an equilateral triangle of height 55 mm and inscribe a circle in it.
4. Inscribe a circle in a given square of side 40 mm.
5. Inscribe a circle in a rhombus whose diagonals are 70 mm and 40 mm.
6. Draw a regular pentagon of side 45 mm and inscribe a circle in it.
7. To draw an arc of an any radius touching two given straight lines at right angles to each other
8. To divide a circle of 25 mm radius into 12 equal parts using compass.
9. To inscribe a regular octagon in a square of 50 mm.
10. To inscribe a regular octagon in a given circle of 30 mm radius.
11. To draw in a 25 mm radius circle, four number of equal circles each touching the given circle and two of the other circles.

INDIAN SCHOOL MUSCAT
SENIOR SECTION
DEPARTMENT OF COMPUTER SCIENCE

CLASS XI COMPUTER SCIENCE (083)
HOLIDAY HOMEWORK
SUBMISSION DATE:26/07/2026

1.	Convert the following decimal numbers into binary: a) 145 b) 78
2.	Convert the following decimal numbers into octal: a) 300 b) 125
3.	Convert the following decimal numbers into hexadecimal: a) 512 b) 1023
4.	Convert the following binary numbers into decimal: a) 1001111 b) 101010
5.	Convert the following octal numbers into decimal a) 57657 b) 2763276
6.	Convert the following hexadecimal numbers into decimal: a) 2F162F b) 7C167C
7.	Simplify and prove using truth table: A+A'B =A+B
8.	Explain and prove following laws using truth table: a) Associative law b) Distributive law
9.	Draw logic gate: Y=((A+B).C)'
10.	Give duals for the following: a) A+(A·B)=A b) A·(B·C)=(A·B)·C

INDIAN SCHOOL MUSCAT
SENIOR SECTION
DEPARTMENT OF COMPUTER SCIENCE

CLASS XI ARTIFICIAL INTELLIGENCE (843)
HOLIDAY HOMEWORK
2026
SUBMISSION DATE: 26, July, 2026

1.	Do an IBM skill build certification course on AI and submit the certificate.
2.	Activity: “Design Your AI Career Path” Part A: AI Career Exploration Write any 5 AI-related careers from the list below and add one line about each: Examples: Machine Learning Engineer Data Scientist AI Ethicist Robotics Engineer NLP Engineer AI Product Manager Computer Vision Engineer Part B: Skill Matching Write at least 3 skills you already have and 3 skills you need to develop. Part C: My AI Future Path Answer the following: Which AI career interests you the most and why? How can AI help society in that field? What steps will you take in the next 2 years to learn AI? (courses, coding, projects, etc.) Part D: Design a “Future Me in AI” poster using any Generative AI tool.

ECONOMICS (030)

TASK 1

QUESTIONNAIRE PREPARATION

Prepare a questionnaire based on the topic: "Income Disparity among People in Your Locality"

Guidelines:

- Prepare a questionnaire with at least 25 questions.
  Include different types of questions:
- Yes/No questions
- Multiple Choice Questions (MCQ)
- Open-ended questions
- Two-option / comparative questions (e.g., high/low income, employed/unemployed)
- Questions should be related to income, occupation, employment, and standard of living.
- Questions must be simple, clear, and suitable for questionnaire format.
- Ensure a balanced mix of all question types.
- Avoid lengthy or complicated questions.
- Maintain neutral and unbiased wording while framing questions.
- Focus on understanding income differences and socio-economic conditions in the locality.

TASK 2

An economy produces two goods – Guns and Butter. The following table shows different production possibilities

Combinations	Guns(units)	Butter(units)
A	0	100
B	1	90
C	2	70
D	3	40
E	4	0

Based on the above table, answer the following questions:

- Construct the Production Possibility Curve (PPC).
- Calculate the Marginal Opportunity Cost (MOC) of producing guns at each level.
- Calculate the Marginal Rate of Transformation (MRT).
- Last date for submission of project is on or before **Sunday, 12th July, 2026.**

PSYCHOLOGY (037)

Project Work Guidelines - 2026-2027

- Title page need to be prepared with the school name, and the topic name and your name and roll number.
- Declaration about the study.
- Acknowledgement (Thanking the teacher, HOD, VP & Principal)
- List of contents.

PHASE – I CHAPTER – 1

1.1 Brief introduction regarding the selected psychological disorder.
1. 2. Characteristics or features associated with the disorder.
1. 3. Causes
1. 4. Historical background
1. 5. Psychologist associated with it
1. 6. Diagnostic procedure
1. 7. Objective of the study (one paragraph)

PHASE – 2 CHAPTER – 2

- Data collection or the presentation of the study (Name of the person, Age, Gender, Birth order, Religion, Nationality, academic background---etc.) need to be followed bio data format.
- Family environment
- Professional life
- Socio economic status
- Diagnosis of the problem
- Observed clinical symptoms.
- Treatment provided to the client. (Including the death of the client if the client passed away)

CHAPTER – 3

- Doctor's report (If available) (One page)
- Individual report of the researcher (Own views about the selected topic and knowledge gained through the project work) (Two pages)
- Suggestions to overcome the above problem.

CHAPTER – 4

- Summary of the case study
- Conclusion (Summary one page and conclusion one page)
- Bibliography

Final submission date for approval is on **13th July, 2026**.

REFERENCES

https://college.mayo.edu/academics/health-sciences-education/ https://www.psychiatry.org/dsm5

https://www.webmd.com/schizophrenia/schizophrenia-research
https://nimhans.ac.in/
https://www.ncbi.nlm.nih.gov/books/NBK558998/ https://bbrfoundation.org/research/bipolar-disorder
https://neura.edu.au/health/mental-health/bipolar-disorder

EXTERNAL ASSESSMENT

TOTAL MARKS: 30

S.No	Title	marks
1.	Project + Viva	15 marks
2.	Lab Journal File	5 marks
3.	Experiment Writing	10 marks